

Federated Computing and Data Infrastructure for Quantitative Research: A Systems-Oriented Review and Reference Architecture

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Abstract

A major opportunity for quantitative research lies in federated systems that turn distributed, institutionally governed data into reliable and verifiable research findings. As quantitative research moves beyond individual price-based signal models toward multimodal, multi-model, and alternative-data research, the opportunity is to coordinate analytics across organisational, contractual, and jurisdictional boundaries while preserving data locality, institutional autonomy, and regulatory compliance.

This paper takes a federated systems view of this opportunity. Building on advances in predictive modelling and federated machine learning, it focuses on the broader architecture required for institution-grade quantitative research: who may request a task, which data and code are authorised, what derived output may leave a node, and how the resulting artefact can be reproduced and challenged. We synthesise literature on financial data assets, AI methods including agentic systems, multimodal and multi-model research, alternative data, data-collaboration architectures, trusted research environments, federated computing, governance, and machine-learning operations, and compare centralised, trusted, and federated designs as complementary responses to different operating conditions.

To make the argument concrete, the paper presents an exemplar reference architecture that separates local collection, policy-aware orchestration, authorised execution, validation and aggregation, and research-facing delivery. Where valuable analysis must cross boundaries that raw data cannot, federated computing provides a foundational pattern for scalable, collaborative, and regulation-aware quantitative research—moving computation to data, and elevating governance and evidence management to first-class system responsibilities.

Keywords: Federated Computing; Quantitative Research; Financial Artificial Intelligence; Distributed Analytics; Alternative Data; Multimodal Learning; Multi-Model Research; Agentic AI; Data Governance; Machine-Learning Operations.

JEL classification: C45; C55; G10; G17; O33.

1. Introduction

Quantitative finance has evolved from narrowly defined statistical and rule-based models operating on structured price and volume data into a research discipline that combines market data, corporate fundamentals, macroeconomic indicators, text, imagery, and other alternative information [1,10,22,37,44].

Advances in machine learning have expanded the analytical toolkit available to researchers, including gradient-boosted trees, deep time-series models, natural-language processing, reinforcement learning, foundation models, and increasingly agentic systems that plan workflows and call tools [11, 19, 27, 39, 41, 43]. The corresponding systems challenge is no longer only how to select a predictive model, but how to acquire, align, govern, reproduce, and evaluate evidence across a complex research lifecycle.

Modern quantitative research is frequently conducted across organisational silos, data-vendor boundaries, cloud and on-premises environments, and multiple legal jurisdictions. Raw data may be commercially licensed, personally sensitive, technically difficult to move, or accessible only for approved purposes. Centralised data platforms remain effective when data can be legally and economically consolidated, but they become less suitable when locality and institutional autonomy are hard constraints. Trusted research environments provide controlled access to sensitive data inside a shared protected environment, while federated computing distributes authorised computation to the locations where data are held [9, 18, 23].

This paper argues that quantitative research should therefore be understood as both a modelling problem and a systems-engineering problem. Predictive quality depends on the surrounding infrastructure: data lineage, temporal alignment, reproducible execution, access control, output validation, model documentation, and the ability to challenge an analytical result. These requirements become more demanding when computation is distributed and when outputs must be assembled without centralising raw data.

The paper makes three contributions. First, it synthesises the data assets and AI method families—including multimodal, multi-model, and agentic capabilities—that shape contemporary quantitative research and translates them into system requirements. Second, it situates federated computing among data-collaboration architectures and compares centralised, trusted-environment, and federated patterns, emphasising the conditions under which each is appropriate and the operational, privacy, and governance trade-offs involved. Third, it presents an exemplar reference architecture that separates local ingestion, policy-aware orchestration, authorised execution, validation and aggregation, provenance, and human-facing delivery. Unlike diagrams focused only on federated model training, the architecture treats governance and evidence management as runtime responsibilities across the full research workflow, without claiming to be a complete Federated Quantitative Intelligence design.

The review is organised around four research questions:

- RQ1.** Which data and AI capabilities shape the infrastructure requirements of modern quantitative research?
- RQ2.** How do centralised platforms, trusted research environments, and federated architectures differ in their treatment of data locality, governance, reproducibility, and operational control?
- RQ3.** Which service boundaries are required to operationalise federated quantitative analytics beyond model training alone?
- RQ4.** What technical, security, and governance gaps remain before federated research architectures can support institution-grade use?

The remainder of the paper proceeds from review method and research workflows to data assets, AI methods, multimodal and multi-model research, data-collaboration architectures, federated services, governance, reference architecture, and open research questions. A companion paper, to be published on SSRN, develops the broader concept of Federated Quantitative Intelligence as a governed operating model for institutional decision support. The present paper stands independently as the systems review

and exemplar architectural foundation; it does not introduce a full multi-agent FQI design, detailed policy engines, interface contracts, or a formal evidence-packaging model.

2. Review Method

This paper adopts a *systems-oriented structured narrative review*. It is not a systematic review, bibliometric study, or meta-analysis. The objective is to integrate evidence from quantitative finance, machine learning, privacy-enhancing technologies, distributed systems, and governance into a coherent architectural account rather than to estimate a pooled empirical effect.

Table 1 summarises how the literature base was assembled and coded. Searches combined terms such as *quantitative research infrastructure, machine learning in finance, alternative data, multimodal financial learning, multi-model research, agentic AI, LLM agents, data collaboration architecture, data mesh, federated computing, federated learning, federated analytics, trusted research environment, secure aggregation, privacy leakage, MLOps, model governance, and data provenance*.

Element	Practice in this review
Sources searched	Scholarly indexes, publisher repositories, major conference proceedings, arXiv, and SSRN
Coverage window	Foundational works retained without hard start year; searches last updated August 2026
Inclusion focus	Substantive contribution to research workflows/data assets, multimodal integration, federated or privacy-preserving computation, MLOps/reproducibility, or governance/provenance/output control
Exclusion focus	Pure market-connectivity or order-routing papers; detailed portfolio construction, order management, and live execution (out of research-infrastructure scope)
Prioritisation	Peer-reviewed research, influential foundations, recent surveys, and authoritative technical or governance frameworks; preprints retained for emerging systems concepts
Working set	49 sources retained in the working bibliography; 45 cited in the main text after relevance screening
Synthesis lenses	Data location; computational responsibility; coordination model; outputs exchanged; governance controls
Limitation	Interpretive and non-exhaustive; the reference architecture synthesises recurring responsibilities rather than proving a single optimal implementation

Table 1: Review protocol for the systems-oriented structured narrative review

Candidate sources were screened for topical relevance against the inclusion categories, then coded against the five architectural lenses to support comparison across communities that use related but non-equivalent terminology. Duplicate or near-duplicate treatments of the same idea were collapsed to the most authoritative or recent representative source. The reference architecture in Section 10 should therefore be read as a synthesis of recurring system responsibilities, not as evidence that one implementation is universally optimal.

3. Quantitative Research Systems: From Models to Platforms

Quantitative research uses mathematical, statistical, and computational methods to generate, test, and interpret hypotheses about financial markets. Earlier workflows were often organised around structured time series, handcrafted factors, and relatively self-contained modelling code [10, 28]. Contemporary research platforms support a broader lifecycle: data discovery and acquisition, quality assessment, feature engineering, exploratory analysis, model development, backtesting, robustness testing, documentation, review, and controlled publication of research artefacts.

The expansion of available data has changed the economics and organisation of this lifecycle. Market and fundamental data are now routinely combined with news, filings, transcripts, web activity, geospatial observations, and ESG information [37, 44]. At the same time, model development has become more dependent on shared feature stores, scalable compute, experiment tracking, versioned datasets, and reproducible pipelines. Machine-learning systems accumulate technical debt through unstable dependencies, hidden feedback loops, data coupling, and changes in upstream inputs [35]. MLOps research consequently treats versioning, orchestration, testing, monitoring, and governance as first-class lifecycle concerns rather than deployment details [25].

Table 2 summarises the shift from isolated modelling workflows to federated research platforms.

Stage	Characteristics	Main limitations
Isolated quantitative models	Structured market data, handcrafted features, local scripts, analyst-specific workflows	Limited reuse, weak lineage, inconsistent reproducibility
Shared research platforms	Common data access, backtesting, notebooks, compute clusters, experiment tracking	Centralisation pressure, platform bottlenecks, organisational silos
AI-enabled research platforms	Machine learning, NLP, alternative data, multimodal and cross-asset signals	High data-engineering burden, model complexity, substantial compute demand
Federated research platforms	Distributed data, authorised local computation, aggregation of derived outputs	Coordination, policy, privacy, reliability, and attribution complexity

Table 2: Evolution of quantitative research systems

A research platform is not merely a larger collection of models. It must preserve the relationship between source data, transformations, assumptions, code, model versions, validation results, and the conclusions presented to users. Dataset documentation and model cards provide useful patterns for making data provenance, intended use, evaluation, and limitations explicit [21, 31]. In institutional environments, these artefacts also support independent review and controlled reuse.

The central architectural question is therefore how to preserve research flexibility while enforcing consistent evidence, reproducibility, and governance. Centralised platforms simplify global visibility and debugging when data movement is permitted. Federated designs become attractive when data cannot be freely consolidated, but they add network, coordination, and trust boundaries that must be designed explicitly.

4. Data Assets for Quantitative Research

Modern quantitative research relies on a spectrum of data assets extending beyond traditional financial time series. Data has become a strategic research input, but its value depends on provenance, permitted use, temporal alignment, quality, and the ability to reproduce transformations applied to it [37,44].

4.1. Structured Financial and Economic Data

The foundation of most quantitative research remains structured market and reference data: historical prices, returns, volumes, bid–ask spreads, order-book observations, instrument identifiers, and corporate actions. These datasets support the construction of signals related to momentum, mean reversion, volatility, liquidity, and market microstructure. Corporate fundamentals, analyst estimates, accounting statements, and valuation measures support factor research and cross-sectional analysis [22].

Macroeconomic and market-wide indicators include interest rates, inflation, employment, commodity prices, surveys, and central-bank communications. Although these datasets are usually structured, they arrive at different frequencies, are subject to revisions, and may require point-in-time reconstruction to avoid look-ahead bias. A research system must therefore retain release timestamps, vintage history, symbology mappings, and transformation lineage rather than only a final analytical table.

4.2. Alternative and Unstructured Data

Alternative data includes non-traditional sources that may provide information about economic activity, corporate behaviour, market attention, or risk. Examples include web traffic, consumer transactions, mobility data, satellite imagery, supply-chain information, and social-media activity [37,44]. Textual sources such as news, company filings, earnings-call transcripts, and research notes are increasingly processed through natural-language models [2,29,42].

ESG data illustrates the governance and quality challenges of alternative information. Coverage, definitions, scoring methods, and missing-data treatment can vary materially across providers [8,40]. Similar issues arise in other alternative datasets where collection methodology may change without notice or where licensing restricts derived outputs and downstream redistribution.

Table 3 summarises the principal data categories.

Data type	Examples	Research uses and system implications
Market data	Prices, returns, order books, volumes, corporate actions	Signal research, volatility and liquidity analysis; requires point-in-time integrity and high-volume storage
Fundamental data	Earnings, balance sheets, cash flow, analyst estimates	Factor construction and valuation; requires entity resolution, restatement handling, and vintage control
Textual data	News, filings, transcripts, research notes	Sentiment, event extraction, semantic retrieval; requires document lineage and model-version tracking
Alternative data	Satellite imagery, web traffic, transactions, mobility	Activity measurement and nowcasting; requires licensing controls, quality monitoring, and specialist preprocessing
ESG data	Disclosures, sustainability metrics, third-party scores	Risk overlays and thematic research; requires uncertainty handling and provider-methodology comparison
Research artefacts	Features, notebooks, models, backtests, reports	Reuse, challenge, and audit; requires versioning, provenance, approvals, and role-scoped access

Table 3: Key data assets and artefacts in quantitative research

4.3. Scale, Heterogeneity, and Temporal Integrity

The integration of diverse financial data introduces five recurring challenges:

- data scale and compute demand;
- heterogeneous formats and representations;
- asynchronous availability, revisions, and market-time alignment;
- missing values, noise, and changing collection processes; and
- contractual, regulatory, and internal-policy restrictions on access and use.

High-frequency datasets can generate very large volumes, while text, images, and geospatial data impose different storage and preprocessing requirements. Multimodal models require feature extraction, timestamp reconciliation, entity resolution, and careful treatment of information availability. These are not neutral engineering details: an apparently strong model may be invalid if it uses revised data, stale identifiers, or information unavailable at the stated research time.

Governance constraints add another dimension. Data may be accessible only in a specific jurisdiction, under a restricted licence, or for an approved research purpose. A federated system must represent these constraints in machine-enforceable policies and carry them into task authorisation and output release. Data locality alone does not ensure compliant use.

5. Artificial Intelligence and Machine Learning in Quantitative Research

Artificial intelligence and machine learning support a wide range of quantitative research tasks, from return prediction and factor discovery to document analysis and scenario generation. Their usefulness

depends not only on predictive performance but also on stability, interpretability, computational cost, and the surrounding controls used to reproduce and challenge results.

5.1. Classical Machine Learning

Random forests, gradient boosting machines, and support-vector machines remain widely used for non-linear factor and ranking problems with comparatively manageable training cost and diagnostics [7,11,22]. Their usefulness still depends on temporal validation, transaction-cost assumptions, and control of multiple testing.

5.2. Deep Learning for Financial Time Series

Recurrent networks and attention-based forecasters, including the Temporal Fusion Transformer, support sequential and multivariate prediction while exposing limited interpretability mechanisms [19,27]. They can encode rich non-linear structure, but remain sensitive to architecture, data scale, and regime change, and therefore require monitoring and robustness testing.

5.3. Natural-Language Processing

Financial news, filings, transcripts, and research documents contain information not represented directly in structured market data. Dictionary-based sentiment methods remain useful as transparent baselines [29]. Domain-adapted transformer models such as FinBERT improve the representation of financial language and support sentiment analysis, event extraction, retrieval, and document classification [2,42].

5.4. Reinforcement Learning and Sequential Research

Reinforcement learning addresses sequential decisions in which actions affect future states and rewards [12,32]. Within this review it is treated as a research method for portfolio simulation and policy evaluation rather than live execution. Performance is highly sensitive to reward design, market simulation, costs, and non-stationarity, so reproducible environments and stress testing are essential.

5.5. Foundation Models and Research Assistants

Foundation models and finance-specific language models support summarisation, document classification, semantic search, extraction, and research assistance [41]. Their ability to combine text with tools and structured data creates opportunities for richer workflows, but also introduces risks of unsupported generation, prompt-based data leakage, non-determinism, and weak source attribution. In research settings, generated conclusions should therefore remain linked to source evidence, model version, prompts or task specifications, and human review.

5.6. Agentic and Autonomous AI

Agentic systems go beyond single-turn generation: they plan multi-step workflows, select and call tools, monitor intermediate outcomes, and refine subsequent actions [39,43]. Relative to static research assistants, they introduce autonomy over task decomposition, tool use, and iteration. Open-source agent

frameworks such as OpenClaw illustrate practical tool-using and workflow-orchestration patterns; the systems requirements that follow do not depend on any single product.

For quantitative research, relevant applications include data discovery and catalogue interrogation, document analysis and event extraction, candidate model or feature selection, monitoring of pipeline health and drift, and draft report or evidence-pack assembly. Multi-agent coordination—in which specialised agents handle retrieval, modelling, validation, or challenge—is a natural extension of multi-model workflows, but it amplifies dependencies among tools, intermediate artefacts, and policy decisions.

These capabilities create distinctive infrastructure and governance requirements. Tool calls must be authorised against data licences, purpose limitation, and node policy; intermediate artefacts need versioning and provenance; and human challenge remains necessary where results inform research conclusions. Principal risks include cascading errors across multi-step plans, unauthorised or over-broad tool use, data leakage through prompts or retrieved context, weak provenance when trajectories are not recorded, excessive autonomy without stop conditions, and unclear accountability for agent-initiated actions. This section therefore treats agentic AI as a literature and systems-requirements concern. Named FQI agent roles, dynamic policy composition, and full multi-agent operating models are left to companion papers.

Method family	Typical models	Research applications and controls
Classical ML	Random forest, XGBoost, SVM	Factor discovery and ranking; requires temporal validation and feature lineage
Deep learning	LSTM, GRU, Transformer, TFT	Time-series and nonlinear representation learning; requires compute, monitoring, and robustness testing
NLP	Sentiment models, FinBERT, domain transformers	News and filing analysis; requires document provenance and language-model versioning
Reinforcement learning	Policy-gradient and actor-critic methods	Sequential allocation research; requires reproducible environments and realistic costs
Foundation models	FinGPT and related LLMs	Summarisation, retrieval, extraction, research assistance; requires evidence links and human challenge
Agentic AI	Tool-using and multi-agent LLM systems	Workflow planning, data discovery, monitoring, report drafting; requires authorised tool use, trajectory provenance, and human oversight

Table 4: AI methods applied in quantitative research

5.7. Why Models Cannot Be Assessed in Isolation

Financial markets are noisy, adaptive, and non-stationary. Complex models can produce unstable conclusions when data distributions, market regimes, vendor methodologies, or feature pipelines change. System quality therefore includes more than model accuracy: it includes data validity, reproducibility, resilience, documentation, monitoring, and the ability to identify which evidence drove a result.

This point also limits claims that more complex models are automatically better. Interpretable models may be preferable in high-stakes settings where a small performance gain does not justify a large reduction in auditability [34]. The appropriate architecture should support multiple model families, transparent baselines, ablation tests, and independent validation rather than embedding one modelling paradigm.

6. Multimodal and Multi-Model Research

Multimodal research combines different forms of data within an analytical workflow, whereas multi-model research combines or coordinates multiple models, services, or specialised pipelines. The concepts are related and often confused, but they create different infrastructure requirements.

6.1. Definitions and Distinction

Concept	What is combined?	Example
Multimodal	Data modalities	Prices + filings + satellite imagery
Multi-model	Models or analytical services	NLP model + forecasting model + risk model
Combined workflow	Both	Multiple models operating across multiple data types

Table 5: Multimodal versus multi-model research

Table 5 summarises the distinction. Multimodal work concerns modality alignment and point-in-time integrity across heterogeneous inputs. Multi-model work concerns model identity, versioning, sequencing, and the comparison or fusion of heterogeneous outputs. Combined workflows inherit both sets of requirements and are increasingly common in institutional research.

6.2. Multimodal Research

Multimodal learning combines structured market and fundamental data with text, images, graphs, or other alternative sources. Research suggests that combining structured and unstructured information can improve forecasting or event interpretation in some settings [4, 13, 42]. Cross-asset methods seek shared representations across equities, fixed income, commodities, currencies, derivatives, or digital assets, reflecting the fact that macroeconomic and information shocks propagate across markets.

The systems requirements are substantial. Inputs may have different frequencies, publication schedules, revision policies, and entity identifiers. Text may refer to an event before structured data records its effect; satellite imagery may be irregular; macroeconomic series may be revised after first publication. A valid research platform must preserve what was known at each historical point and distinguish observation time, event time, processing time, and publication time.

Key challenges include alignment of asynchronous signals and point-in-time availability; reconciliation of frequencies, entity mappings, and coverage; interpretability and performance attribution across modalities; changing model and data dependencies; computational cost; and licensing constraints on combined features and derived outputs. Multimodal research therefore strengthens the case for modular infrastructure and can motivate federation when specialist datasets must remain in controlled environments. Semantic alignment still requires shared schemas, metadata, evaluation protocols, and provenance.

6.3. Multi-Model Research

Multi-model research uses multiple models, analytical services, or specialised pipelines to examine the same research question. These may include models designed for different datasets, asset classes, prediction horizons, market regimes, or analytical tasks.

A multi-model workflow may take several forms. Models may operate independently and produce results that are compared by a researcher; their predictions may be combined through an ensemble or weighting mechanism; or they may be arranged sequentially, with one model producing features or evidence for another. For example, a language model may extract events from corporate disclosures, a time-series model may estimate their expected market effect, and a risk model may evaluate whether the resulting signal remains valid under changing volatility conditions. Multi-model analysis can therefore improve robustness and provide alternative perspectives, but it also increases dependencies among data, models, assumptions, and intermediate outputs.

The supporting infrastructure must retain the identity, version, configuration, training context, and intended use of every participating model. It must also record dependencies between models and preserve the lineage of intermediate features, predictions, evidence, and validation results. Where models disagree, the system should expose that disagreement rather than hiding it within a single aggregated result.

6.4. Combined and Federated Workflows

In practice, multimodal and multi-model designs often co-occur: multiple specialised models operate over multiple data types. Federated environments intensify the challenge because participating models may execute across autonomous nodes with different data distributions, infrastructure, policies, and permitted outputs. A federated multi-model system must determine which models are eligible to run at each node, whether model code or parameters may cross institutional boundaries, and how heterogeneous outputs should be compared or combined. Aggregation does not necessarily mean averaging model parameters. Depending on the workflow, it may involve prediction ensembles, ranking, calibration, evidence fusion, model selection, or the presentation of separate node-level results for human comparison.

Policy-aware orchestration is therefore required to select compatible models and execution nodes, while validation services must assess schema compatibility, temporal consistency, calibration, uncertainty, and potential duplication of evidence. Local execution enables specialised models to operate close to governed data; orchestration coordinates model selection and sequencing; validation and aggregation determine how outputs may be compared or combined; and provenance services retain the evidence required to reproduce and challenge the final result. In this setting, the objective is not simply to produce one global model, but to create a governed analytical workflow in which multiple models can contribute traceable and complementary evidence.

7. Data Collaboration Architectures for Quantitative Research

Quantitative research that spans teams, vendors, or jurisdictions must choose how data, computation, and authority are organised. Federation is one option among several collaboration architectures. This section therefore begins with architectural definitions and comparison, then sets out the conditions under which centralised, trusted, or federated designs are appropriate, and only then develops federated computing in the detail needed for the services in Section 8.

7.1. Architectural Definitions

Institutions commonly encounter six related patterns. A *centralised database or data lake* consolidates datasets into a shared store for common tooling and exploratory analysis. A *trusted research environment* (TRE) admits researchers or approved code into a controlled shared boundary and releases only

disclosure-reviewed outputs [9, 23]. A *data mesh* decentralises ownership of governed data products while emphasising discoverability and domain accountability. *Distributed computing* partitions data and tasks across machines primarily for scale, throughput, or resilience, without necessarily preserving institutional data sovereignty. *Federated learning* and *federated analytics* keep raw records local while exchanging model updates or query results [15, 30]. *Federated computing* is broader still: it coordinates authorised services and workflows across autonomous environments under sovereignty and policy constraints [18].

Tables 6 and 7 locate these options relative to one another.

Architecture type	Strengths	Limitations
Shared database or data lake	Central visibility, simpler exploratory analysis, easier debugging, common tooling, lower coordination overhead	Raw-data centralisation, scaling pressure, replication cost, and difficult cross-jurisdiction governance
Trusted research environment	Strong local controls, protected access, disclosure review, reproducible contained environment	Often limited to one protected boundary; code and output release can be slow; multi-partner coordination remains difficult
Federated computing	Data locality, institutional autonomy, collaborative analytics, reduced raw-data movement	Complex orchestration, semantic alignment, communication, output control, failure handling, and multi-party accountability

Table 6: Centralised and distributed architectures for multi-dataset research

Concept	What is coordinated or moved?	Primary objective	Typical output
Distributed computing	Data and tasks may be partitioned across machines	Scale, throughput, resilience, or latency	Computation result
Data mesh	Ownership and publication of decentralised data products	Organisational scalability and domain accountability	Discoverable governed data products
Trusted research environment	Researchers or code enter a controlled shared environment	Secure analysis of sensitive data within a protected boundary	Disclosure-controlled outputs
Federated learning	Model code and parameter updates; raw training data remains local	Collaborative model training across decentralised data	Global or personalised model
Federated analytics	Queries or analytical tasks; raw records remain local	Cross-node statistics and descriptive analysis	Aggregated statistics or reports
Federated computing	Authorised services and workflows across autonomous environments	Broad cross-boundary analytics under sovereignty constraints	Models, statistics, features, evidence, or research artefacts

Table 7: Distinguishing data-collaboration concepts used in quantitative research

7.2. Choosing between Centralised, Trusted, and Federated Designs

Federation should not be treated as universally superior. Selection depends on operating constraints rather than on modelling fashion. Useful questions include:

- Can raw data legally and contractually move into a shared store?
- Is shared infrastructure acceptable to the participating institutions?
- Are disclosure controls required for outputs leaving a protected boundary?
- Must institutions retain local authority over data, code, and release decisions?
- Is the analytical value of cross-boundary collaboration sufficient to justify federation's coordination cost?

When data can be lawfully and economically consolidated, a central platform may provide lower latency, simpler debugging, stronger reproducibility, and lower operational overhead. A TRE is often preferable when sensitive data can be analysed inside one protected environment but cannot be broadly redistributed. Federation becomes most valuable when locality, licensing, sovereignty, or organisational autonomy are binding requirements and when collaboration across those boundaries is analytically worthwhile. The three designs are therefore complementary responses to different conditions, not successive stages of a single maturity path.

7.3. Federated Computing for Quantitative Research

Federated computing coordinates authorised computation across independent environments while preserving local control of data and infrastructure [18]. Depending on the workflow, nodes may execute statistical queries, feature extraction, model training, inference, backtests, data-quality checks, or document analysis and return only approved derived outputs.

A typical federated system includes local data and compute nodes, a coordination or orchestration layer, output validation and aggregation, and cross-cutting identity, policy, provenance, and monitoring services. The architectural objective is not simply to avoid moving raw data, but to control what computation may run, what evidence it may use, what output may leave a node, and how the resulting artefact can be reproduced and challenged.

Federated learning is a well-developed subset of this design space. Foundational work defines collaborative training over decentralised data, while later surveys identify challenges involving non-identically distributed data, communication, privacy, reliability, and fairness [24, 26, 30, 38]. Federated analytics extends the model to statistics and queries rather than model parameters alone [15]. Production systems also show the importance of selection, scheduling, fault tolerance, and secure aggregation [5, 6]. These foundations motivate the service boundaries developed in the next section: local collection, policy-aware orchestration, authorised execution, validation and aggregation, provenance, and research delivery.

8. Federated Services for Quantitative Research

The collaboration architectures introduced in Section 7 become operational only when they are expressed as services that support specific quantitative-research workflows. A federated platform must do more

than distribute model training: it must coordinate data discovery, factor computation, benchmarking, backtesting, document analysis, validation, and evidence delivery while preserving the authority of each participating institution. This section grounds the core service responsibilities in representative quantitative-research examples.

8.1. Local Collection and Registration

Each participating node acquires and maintains data within its own institutional boundary. A collector may ingest market prices, company fundamentals, analyst research, alternative datasets, or existing model artefacts, while a local catalogue records metadata such as ownership, licence conditions, coverage, quality, point-in-time availability, and permitted use.

Consider a transaction dataset licensed for internal factor research but prohibited from being copied into a shared research platform. The local collector normalises the records, links them to approved security identifiers, and registers the dataset’s historical coverage and licence restrictions. Other nodes may discover that the dataset exists and understand its permitted analytical uses, but they do not receive its raw records.

Local registration is therefore not only a data-engineering function. It provides the information required for later decisions about whether a proposed task is technically possible, temporally valid, and contractually permitted.

8.2. Policy-Aware Orchestration

The federated orchestrator receives a research-task specification and determines which nodes are potentially able to participate. The specification should describe the requested analysis, required data characteristics, approved code or model, intended purpose, expected output, and relevant time horizon.

For example, a researcher may request a comparison of a value factor across equity datasets held by several regional teams. The orchestrator identifies nodes containing the required fundamentals and price histories, verifies that their metadata satisfy the requested dates and markets, and dispatches the task only to eligible nodes.

Eligibility does not imply authorisation. Each node retains the authority to accept, modify, or reject the task according to its own licences, jurisdictional obligations, resource constraints, and institutional policy. The orchestrator coordinates the workflow but does not override local control or receive the underlying raw data.

8.3. Authorised Local Execution

Authorised execution means that approved analytical code runs close to the data under node-specific controls. The meaning of authorisation depends on the workflow.

In a restricted-dataset factor benchmark, a node may permit an approved factor implementation to run against locally licensed transaction or alternative data. The execution environment verifies the code version, requested variables, research purpose, temporal cut-off, and permitted output schema. It may return factor returns, coverage statistics, turnover, information coefficients, or uncertainty measures, while prohibiting row-level observations or features that could reconstruct the licensed dataset.

A distributed backtest requires different controls. Each node may execute the same strategy against its own market, asset class, or historical dataset using a pinned portfolio-construction method, transaction-cost

model, rebalancing schedule, and point-in-time universe. The task must specify these assumptions explicitly so that differences in results can be attributed to data or market conditions rather than inconsistent implementations.

Authorised execution may also include local document extraction, data-quality checks, model inference, stress testing, or feature computation. In each case, the node decides which data and code may be combined, what resources may be used, and which outputs are eligible for release.

8.4. Output Validation and Aggregation

Outputs should not leave a node merely because execution completed successfully. Local and shared validation services must check that results conform to the expected schema, quality thresholds, disclosure rules, and permitted level of detail.

For the restricted factor benchmark, validation may confirm that the sample is sufficiently large, that no security-level records are exposed, and that released statistics cannot reveal proprietary observations. Results from participating nodes may then be compared or combined into aggregate performance and coverage measures.

For the distributed backtest, aggregation does not necessarily mean averaging returns. The system may instead align results by market calendar, compare risk-adjusted performance across nodes, construct a combined portfolio simulation, or preserve separate node-level results for attribution and challenge. The aggregation method must therefore be part of the task specification and appropriate to the analytical meaning of the outputs.

Other workflows may require evidence packaging rather than mathematical aggregation. A document-analysis task, for example, may return approved extracted events with source references, confidence scores, and model-version information for later comparison.

8.5. Artefact and Provenance Management

A federated result must remain traceable even when underlying data cannot be exported. The platform should retain enough provenance to reconstruct what was requested, authorised, executed, validated, and released: identities and purpose, code and model versions, configuration and environment references, data-version identifiers, policy decisions, validation outcomes, and approvals.

For a distributed backtest, this includes the security universe, data vintage, feature definitions, cost assumptions, seeds, and participating nodes; for a restricted-data benchmark, the dataset version and disclosure controls, without exposing protected data. Because some evidence must remain local, shared provenance may hold approved metadata and released artefacts rather than complete local code, models, or datasets—enabling verification and challenge without undermining sovereignty.

8.6. Research Delivery and Human Challenge

Federated services ultimately support research rather than merely completing distributed jobs. Research-facing interfaces should allow authorised users to submit workflows, inspect participating nodes, compare outputs, review uncertainty, trace results to evidence, and request further validation.

A reviewer of a cross-node factor benchmark should be able to see whether a strong aggregate result is driven by one market, provider, or modelling assumption; a reviewer of a distributed backtest should be able to challenge inconsistent cost assumptions, stale vintages, failed nodes, or incomplete coverage

and then submit a revised task. The architecture must support this inspect–revise–approve loop while retaining the evidence and decisions for each iteration.

Table 8 summarises these responsibilities for the two workflows. Figure 1 and Section 10 show where the same responsibilities sit technically.

Core responsibility	Restricted-dataset factor benchmark	Distributed backtest	Evidence and controls
Local collection and registration	Register licensed dataset, coverage, identifiers and permitted uses	Register point-in-time prices, universes and cost data	Licence, lineage, quality and vintage metadata
Policy-aware orchestration	Select nodes authorised to benchmark the factor	Select nodes with compatible markets, histories and compute	Identity, purpose, eligibility and node decisions
Authorised execution	Run approved factor code; prohibit row-level output	Run pinned strategy, costs, calendar and rebalancing rules	Code/model version, configuration and environment
Output validation	Check minimum sample size and disclosure risk	Check completion, comparable assumptions and valid periods	Schema, quality, privacy and release decisions
Aggregation or comparison	Combine summary statistics or compare node results	Align returns, compare risks or simulate a combined portfolio	Aggregation method, participating nodes and exclusions
Provenance and delivery	Present factor performance with coverage and uncertainty	Present results with costs, failures and node attribution	Provenance records, approvals and challenge history

Table 8: Quantitative-research examples of core federated service responsibilities

9. Governance, Security, and Trust in Federated Research

Governance in federated research is a runtime systems concern rather than a post-processing review. The following questions acknowledge the conceptual influence of the Federated Computing and FCaC governance grammar [16–18]: who may request a task, which data and code are authorised, what output may leave a node, and how the resulting artefact can be reproduced, challenged, and approved. The architecture must therefore carry policy and evidence through the workflow.

Governance requirements can be grouped into six categories:

- **data sovereignty and purpose limitation:** preserve ownership, licence, jurisdiction, and approved-use constraints;
- **identity and authorisation:** authenticate users, services, and nodes and enforce least-privilege access;
- **provenance and reproducibility:** link outputs to data versions, code, configuration, models, and execution records;
- **model and analytical transparency:** document intended use, assumptions, evaluation, limitations, and known failure modes [21, 31];
- **auditability and accountability:** retain tamper-evident records of requests, approvals, policy decisions, execution, and release; and
- **operational resilience:** monitor node health, failures, drift, communication, and policy violations.

The NIST AI Risk Management Framework provides a useful high-level structure around governing, mapping, measuring, and managing AI risks [33]. In financial institutions, model-risk controls and independent validation also require evidence that results are reproducible, limitations are known, and material changes are reviewed. The architecture should support those controls without assuming that a single global policy can override local obligations.

9.1. Privacy and Security Failure Modes

Keeping raw data local reduces some exposure but does not itself guarantee privacy or security. Shared model updates may leak training information through gradient reconstruction [45]; model outputs can support membership inference [36]; and malicious participants may poison or backdoor a collaboratively trained model [3]. Derived research artefacts can also reveal sensitive features, licensed content, or proprietary methodology even when they contain no raw records.

Relevant controls include secure aggregation, differential privacy, confidential computing, signed and attested workloads, sandboxed execution, output disclosure review, anomaly detection, robust aggregation, and node-level rate or query controls [6, 14, 20]. These controls introduce trade-offs in accuracy, cost, latency, debuggability, and operational complexity. They should therefore be selected through threat modelling rather than applied as generic labels.

Failure mode	Research impact	Illustrative controls
Gradient or output leakage	Exposure of sensitive or proprietary local information	Secure aggregation, differential privacy, disclosure testing, output minimisation
Poisoned or compromised node	Corrupted global model, metrics, or research conclusions	Attestation, signed artefacts, robust aggregation, anomaly detection, quarantine
Schema and semantic mismatch	Invalid comparison or aggregation across nodes	Shared schemas, metadata validation, and calibration checks
Non-IID data and selection bias	Poor global performance or misleading attribution	Node-level diagnostics, stratified evaluation, personalised or hierarchical models
Partial failure and stale nodes	Incomplete or irreproducible workflows	Version pinning, recovery procedures, and explicit handling of incomplete participation
Policy conflict	Unauthorised data use or output release	Local policy enforcement and local authorisation of use and release

Table 9: Principal failure modes and controls in federated quantitative research

10. Federated AI System Architectures

Relative to conventional cloud AI platforms, a federated design assumes autonomous participating environments that may differ in infrastructure, data semantics, policy, and availability. Negotiation and validation therefore occur at boundaries rather than inside a uniform internal environment. Table 10 summarises the recurring layers.

Layer	Role	Typical examples
Local data and meta-data	Distributed financial and alternative datasets with lineage and policy	Catalogues, governed datasets, and data products
Local compute	Isolated execution close to data	Isolated runtimes and protected execution environments
Research services	Feature engineering, training, inference, statistics, backtests, document analysis	Forecasting models, NLP pipelines, factor analysis
Orchestration	Policy-aware workflow coordination across nodes	Task selection, dispatch, and workflow state across nodes
Validation and aggregation	Check and combine authorised local outputs	Secure aggregation, statistics merging, signal fusion, disclosure checks
Provenance and monitoring	Traceability, reproducibility, reliability, and security	Logs, metrics, lineage records, and operational alerts
Research delivery	Human interaction and controlled publication	Workflow interfaces, notebooks, evidence views, and approval queues

Table 10: Typical layers of federated quantitative research architectures

10.1. An Exemplar Federated Quantitative Research Architecture

Figure 1 presents an exemplar reference architecture. It identifies recurring responsibilities rather than prescribing a complete Federated Quantitative Intelligence system, interface contracts, or a specific deployment topology. The figure distinguishes data-ingestion flows, control flows, and approved-output flows. Each node contains its own collector, local data assets, policy enforcement, and execution environment. The orchestrator coordinates task specifications rather than raw datasets. At a high level, a governance and trust plane applies across orchestration, execution, aggregation, and delivery.

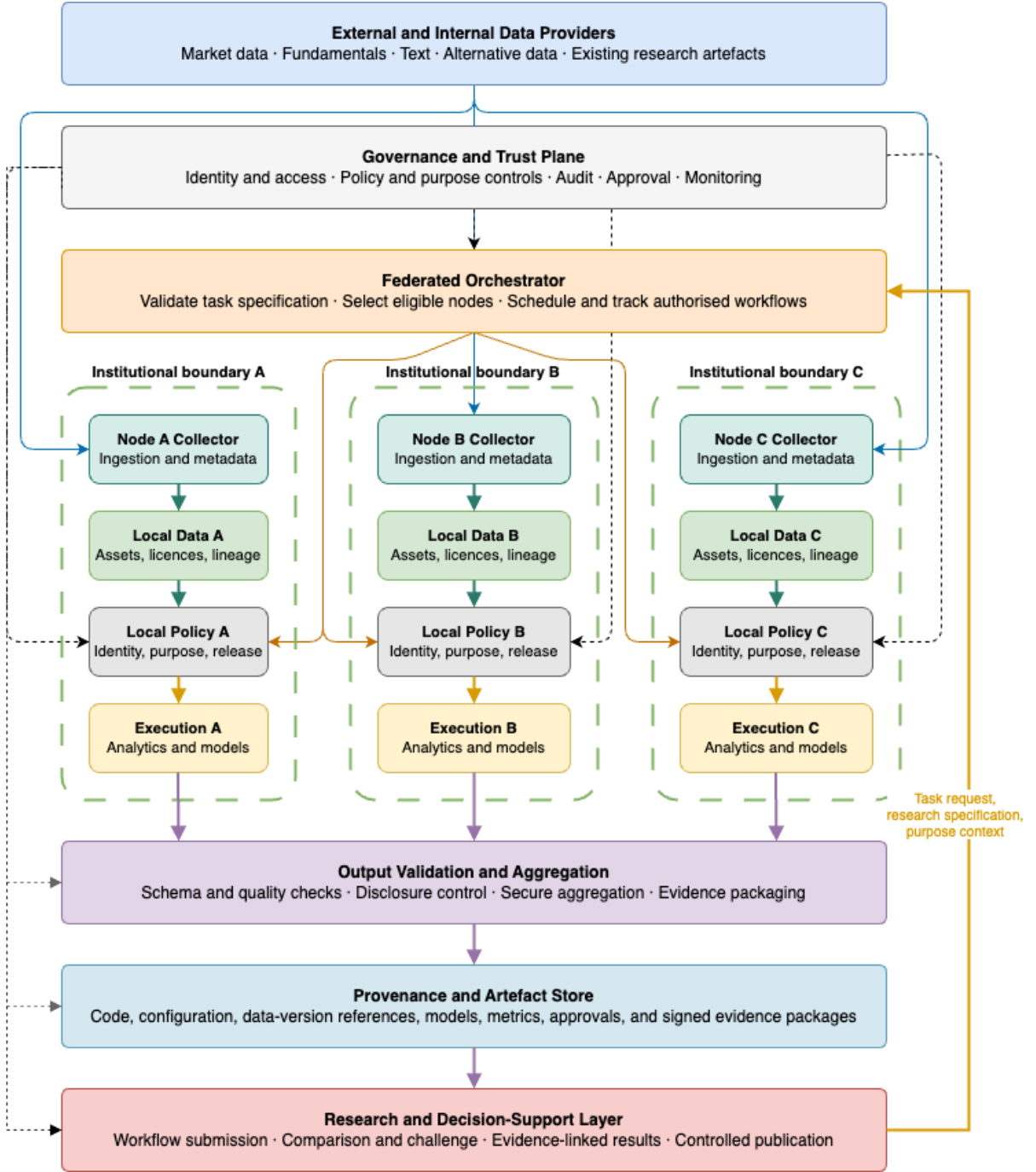


Figure 1: Exemplar reference architecture for federated quantitative research.

Raw data remains within each institutional node; the orchestrator distributes authorised task specifications, and only validated derived outputs are aggregated and delivered.

The component roles in Figure 1 realise the service responsibilities illustrated for quantitative workflows in Table 8. Table 10 provides the corresponding layer legend.

The exemplar therefore makes four design commitments explicit: collectors and policy gates sit inside each institutional boundary; control traffic is separated from data and output flows; aggregation is paired with validation and provenance rather than treated as a neutral mathematical step; and governance is applied at points of action. Detailed policy composition, named agents, service contracts, and formal evidence packaging are left to later work. The architecture ends at evidence-linked research and decision

support. Portfolio construction, pre-trade controls, order management, market connectivity, and post-trade processing may consume approved research outputs, but they introduce additional safety, latency, and regulatory requirements beyond this review.

11. Open Challenges and Research Directions

Institution-grade quantitative research still outruns current federated-learning and distributed-analytics practice. Priority gaps include scalability and communication overhead; semantic interoperability across identifiers, calendars, and vintages; non-stationarity under node heterogeneity; multimodal and multi-model attribution; governed tool use and accountability for agentic workflows; privacy–utility trade-offs; distributed reproducibility when data cannot be exported; robustness to malicious or faulty participants; high-level policy alignment across licences and jurisdictions; and interfaces that support human challenge rather than passive acceptance of outputs.

Evaluation should therefore report more than predictive accuracy: end-to-end latency, communication volume, failure tolerance, reproducibility, policy consistency, privacy leakage, provenance completeness, and time-to-challenge, always against a centralised baseline. Without that baseline, the cost of federation cannot be justified. Companion Federated Quantitative Intelligence work can extend this systems foundation toward a broader operating model and dynamic governance; those developments lie beyond the exemplar architecture presented here. The open empirical question is whether such an operating model improves useful, traceable collaboration without creating unacceptable complexity or false confidence.

12. Conclusion

This review examined the data assets, AI methods, distributed services, governance controls, and architectural patterns shaping modern quantitative research. In response to **RQ1**, the literature shows that heterogeneous and multimodal data, multi-model workflows, and emerging agentic systems make point-in-time integrity, metadata, provenance, authorised tool use, scalable compute, and model lifecycle controls central research requirements. In response to **RQ2**, centralised platforms remain preferable where consolidation is permitted and operational simplicity dominates, whereas trusted environments and federated architectures address stronger locality and sovereignty constraints at the cost of coordination complexity.

In response to **RQ3**, federated quantitative research requires more than decentralised model training. It requires local collection and metadata, policy-aware orchestration, authorised execution, output validation, aggregation, provenance, monitoring, and role-scoped delivery. In response to **RQ4**, major gaps remain in semantic interoperability, privacy leakage, malicious-node resilience, distributed reproducibility, cross-boundary policy alignment, and human challenge.

The resulting conclusion is deliberately conditional. Federated computing is not a default replacement for centralised research platforms. It is an architectural response to situations in which valuable analysis spans data or institutions that cannot be freely centralised. Under those conditions, the effectiveness of the research environment depends as much on governance, evidence, and system design as on the predictive models it contains.

Declarations

Author Contributions. Ethan Nguyen led the conceptualisation of the paper, developed the exemplar reference architecture, prepared the visualisations, and wrote the manuscript. Philip Treleaven contributed to conceptual framing, supervision, technical review, and manuscript revision. Both authors reviewed and approved the final manuscript.

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Ethics Statement. This research did not involve human participants, personal data, live trading activity, or proprietary market operations. No ethics-committee approval was required.

Declaration of Generative AI Use. During preparation of this manuscript, the authors used generative AI tools for language refinement, structural editing, and assistance in identifying areas requiring additional literature review. The authors reviewed and revised all generated material and take responsibility for the manuscript’s content, references, analysis, and conclusions.

References

- [1] Irene Aldridge. *High-Frequency Trading: A Practical Guide to Algorithmic Strategies and Trading Systems*. Wiley, 2013.
- [2] Dogu Araci. FinBERT: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*, 2019.
- [3] Eugene Bagdasaryan, Andreas Veit, Yiqing Hua, Deborah Estrin, and Vitaly Shmatikov. How to backdoor federated learning. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pages 2938–2948, 2020.
- [4] Tadas Baltrušaitis, Chaitanya Ahuja, and Louis-Philippe Morency. Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(2):423–443, 2019.

- [5] Keith Bonawitz, Hubert Eichner, Wolfgang Grieskamp, Dzmitry Huba, Alex Ingerman, Vladimir Ivanov, Chloe Kiddon, Jakub Konečný, Stefano Mazzocchi, H. Brendan McMahan, et al. Towards federated learning at scale: System design. *Proceedings of Machine Learning and Systems*, 1:374–388, 2019.
- [6] Keith Bonawitz, Vladimir Ivanov, Ben Kreuter, Antonio Marcedone, H. Brendan McMahan, Sarvar Patel, Daniel Ramage, Aaron Segal, and Karn Seth. Practical secure aggregation for privacy-preserving machine learning. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, pages 1175–1191, 2017.
- [7] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.
- [8] Sergio Caprioli, Jacopo Foschi, Riccardo Crupi, and Alessandro Sabatino. Denoising ESG: Quantifying data uncertainty from missing data with machine learning. *arXiv preprint arXiv:2407.20047*, 2024.
- [9] Ed Chalkrey. Developing and publishing code for trusted research environments: Best practices and ways of working. *arXiv preprint arXiv:2111.06301*, 2021.
- [10] Ernest P. Chan. *Algorithmic Trading: Winning Strategies and Their Rationale*. Wiley, 2012.
- [11] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.
- [12] Yue Deng, Fuchun Bao, Youyong Kong, Zhiqiang Ren, and Qionghai Dai. Deep direct reinforcement learning for financial signal representation and trading. *IEEE Transactions on Neural Networks and Learning Systems*, 28(3):653–664, 2016.
- [13] Xiao Ding, Yue Zhang, Ting Liu, and Junwen Duan. Deep learning for event-driven stock prediction. In *Proceedings of IJCAI*, pages 2327–2333, 2015.
- [14] Cynthia Dwork and Aaron Roth. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9(3–4):211–407, 2014.
- [15] Ahmed Roushdy Elkordy, Yahya H. Ezzeldin, Shanshan Han, Shantanu Sharma, Chaoyang He, Sharad Mehrotra, and Salman Avestimehr. Federated analytics: A survey. *arXiv preprint arXiv:2302.01326*, 2023.
- [16] Enzo Fenoglio and Philip Treleaven. Auditable session admission for cross-silo federated learning. In *2026 2nd International Conference on Federated Learning and Intelligent Computing Systems (FLICS)*, pages 231–238. IEEE, 2026. doi:[10.1109/FLICS70075.2026.11621917](https://doi.org/10.1109/FLICS70075.2026.11621917).
- [17] Enzo Fenoglio and Philip Treleaven. Federated computing as code (FCaC): Sovereignty-aware systems by design. *arXiv preprint arXiv:2603.17331*, 2026.
- [18] Enzo Fenoglio and Philip Treleaven. Federated computing: information integration under sovereignty constraints. *Royal Society Open Science*, 13(2):251318, 2026.
- [19] Thomas Fischer and Christopher Krauss. Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2):654–669, 2018.

- [20] David Froelicher, Juan R. Troncoso-Pastoriza, Jean Louis Raisaro, Michel A. Cuendet, João S. Sousa, Hyunghoon Cho, Bonnie Berger, Jacques Fellay, and Jean-Pierre Hubaux. Truly privacy-preserving federated analytics for precision medicine with multiparty homomorphic encryption. *Nature Communications*, 12:5910, 2021.
- [21] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021.
- [22] Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020.
- [23] Emily Jefferson, James Liley, Maeve Malone, Smarti Reel, Alba Crespi-Boixader, Xaroula Kerassidou, Francesco Tava, Andrew McCarthy, Richard Preen, Alberto Blanco-Justicia, et al. GRAIMAT-TER green paper: Recommendations for disclosure control of trained machine learning models from trusted research environments. *arXiv preprint arXiv:2211.01656*, 2022.
- [24] Peter Kairouz, H. Brendan McMahan, Brendan Avent, Aurélien Bellet, Mehdi Bennis, Arjun Nitin Bhagoji, Kallista Bonawitz, Zachary Charles, Graham Cormode, Rachel Cummings, et al. Advances and open problems in federated learning. *Foundations and Trends in Machine Learning*, 14(1–2):1–210, 2021.
- [25] Dominik Kreuzberger, Niklas Kühl, and Sebastian Hirschl. Machine learning operations (MLOps): Overview, definition, and architecture. *IEEE Access*, 11:31866–31879, 2023.
- [26] Tian Li, Anit Kumar Sahu, Ameet Talwalkar, and Virginia Smith. Federated learning: Challenges, methods, and future directions. *IEEE Signal Processing Magazine*, 37(3):50–60, 2020.
- [27] Bryan Lim, Sercan O. Arik, Nicolas Loeff, and Tomas Pfister. Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4):1748–1764, 2021.
- [28] Andrew W. Lo. Machine learning in finance. In *Research Handbook on Big Data and Law*. Edward Elgar Publishing, 2017.
- [29] Tim Loughran and Bill McDonald. When is a liability not a liability? textual analysis, dictionaries, and 10-ks. *The Journal of Finance*, 66(1):35–65, 2011.
- [30] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, pages 1273–1282, 2017.
- [31] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, pages 220–229, 2019.
- [32] John Moody and Matthew Saffell. Learning to trade via direct reinforcement. *IEEE Transactions on Neural Networks*, 12(4):875–889, 2001.
- [33] National Institute of Standards and Technology. Artificial intelligence risk management framework (AI RMF 1.0). Technical Report NIST AI 100-1, U.S. Department of Commerce, 2023.

- [34] Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1:206–215, 2019.
- [35] D. Sculley, Gary Holt, Daniel Golovin, Eugene Davydov, Todd Phillips, Dietmar Ebner, Vinay Chaudhary, Michael Young, Jean-François Crespo, and Dan Dennison. Hidden technical debt in machine learning systems. In *Advances in Neural Information Processing Systems*, volume 28, 2015.
- [36] Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *2017 IEEE Symposium on Security and Privacy*, pages 3–18, 2017.
- [37] Yunchuan Sun, Lu Liu, Ying Xu, et al. Alternative data in finance and business: Emerging applications and theory analysis. *Financial Innovation*, 10:127, 2024.
- [38] Jianyu Wang, Zachary Charles, Zheng Xu, Gauri Joshi, H. Brendan McMahan, Blaise Agüera y Arcas, Maruan Al-Shedivat, Katharine Daly, Deepesh Data, Suhas Diggavi, et al. A field guide to federated optimization. *arXiv preprint arXiv:2107.06917*, 2021.
- [39] Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhiyuan Chen, Jiakai Tang, Xu Chen, Yankai Lin, Wayne Xin Zhao, Zhe Wei, and Ji-Rong Wen. A survey on large language model based autonomous agents. *Frontiers of Computer Science*, 18(6):186345, 2024.
- [40] Jun Xu. AI in ESG for financial institutions: An industrial survey. *arXiv preprint arXiv:2403.05541*, 2024.
- [41] Xiao-Yang Yang et al. FinGPT: Open-source financial large language models. *arXiv preprint arXiv:2306.06031*, 2023.
- [42] Yi Yang et al. FinBERT: A pretrained language model for financial communications. *arXiv preprint arXiv:2006.08097*, 2020.
- [43] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023.
- [44] Qingquan Tony Zhang, Beibei Li, and Danxia Xie. *Alternative Data and Artificial Intelligence Techniques: Applications in Investment and Risk Management*. Springer, 2022.
- [45] Ligeng Zhu, Zhijian Liu, and Song Han. Deep leakage from gradients. In *Advances in Neural Information Processing Systems*, volume 32, 2019.